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Strong Discrete Morse Theory and Simplicial L–S Category: A Discrete Version of the Lusternik–Schnirelmann Theorem

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Abstract
We prove a discrete version of the Lusternik–Schnirelmann (L–S) theorem for discrete Morse functions and the recently introduced simplicial L–S category of a simplicial complex. To accomplish this, a new notion of critical object of a discrete Morse function is presented, which generalizes the usual concept of critical simplex (in the sense of R. Forman). We show that the non-existence of such critical objects guarantees the strong homotopy equivalence (in the Barmak and Minian's sense) between the corresponding sublevel complexes. Finally, we establish that the number of critical objects of a discrete Morse function defined on K is an upper bound for the non-normalized simplicial L–S category of K.

Keywords Simplicial Lusternik–Schnirelmann category · Strong collapsibility · Discrete Morse theory · Strong homotopy type

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1 Introduction

Since its inception by Forman [10], discrete Morse theory has been a powerful and versatile tool used not only in diverse fields of mathematics, but also in applications to other areas [15] as well as a computational tool [7]. Its adaptability stems in part from the fact that it is a discrete version of the beautiful and successful "smooth" Morse theory [12]. While smooth Morse theory touches many branches of mathematics, one such area that has its origins in critical point theory is that of Lusternik–Schnirelmann category. The (smooth) Lusternik–Schnirelmann category or L–S category of a smooth manifold X, denoted $\operatorname{cat}(X)$, was first introduced in [11], where the authors proved what is now known as the L–S Theorem (see also [6] for a detailed survey of the topic). A version of this result can be stated as follows:

Theorem 1.1 *If M is a compact smooth manifold and $f : M \to \mathbb{R}$ is a smooth map, then*

$$\operatorname{cat}(M) + 1 \leq \#\operatorname{crit}(f), \tag{1}$$

where $\operatorname{crit}(f)$ *is the set of all critical points of the function f.*

There are many "smooth" versions of this theorem in various contexts (see for example [14]). The aim of this paper is to view Forman's discrete Morse theory from a different perspective in order to prove a discrete version of the L–S theorem compatible with the recently defined simplicial L–S category developed by three of the authors [9]. This simplicial version of L–S category is suitable for simplicial complexes. Other attempts have been made to develop a version of L–S category adapted to the simplicial setting. In [1], one of the authors developed a so-called *discrete* L–S category and proved an analogous L–S theorem for discrete Morse functions. One major drawback, however, is that the discrete L–S category is not an invariant under any kind of reasonable notion of equivalence and hence its usefulness appears to be quite limited. Two similar combinatorial approaches were attempted by Tanaka [17]. In contrast, the simplicial category is known to be an invariant of the strong homotopy type of a simplicial complex (Theorem 4.4). It furthermore generalizes to simplicial maps and enjoys many of those same properties shared by the classical version [16]. Thus a simplicial version of the L–S theorem seems appropriate. Our version, Theorem 4.5, relates a new generalized notion of critical object of a discrete Morse function to the simplicial L–S category of [9].

In this paper we use the notion of simplicial L–S category defined in [9]. This simplicial approach of L–S category uses strong collapses in the sense of Barmak and Minian [3] as a framework for the notion of categorical subcomplexes. Instead of the classical notion of collapse, it is natural to consider strong collapses in the simplicial setting since they correspond to simplicial maps (see Fig. 1).

This is especially contrasted with (classical) elementary collapses, the basic moves of the simple homotopy type theory introduced by Whitehead [5], which in general do not correspond to simplicial maps (see Fig. 2).

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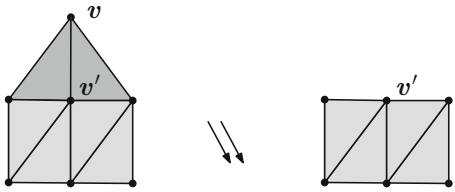

Fig. 1 An elementary strong collapse from K to $K - \{v\}$

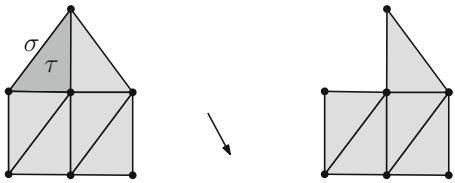

Fig. 2 An elementary (classical) collapse from K to $K - \{\sigma, \tau\}$

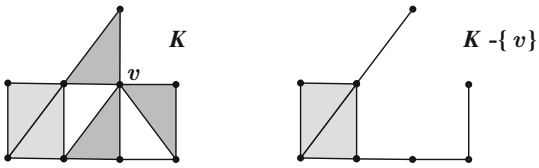

Fig. 3 Deletion of a vertex that does not correspond to a simplicial map from K to $K - \{v\}$

Furthermore, elementary strong collapses correspond to the deletion of the open star of a dominated vertex. Notice that, if a vertex is not dominated by another one, the deletion of its open star is not a simplicial map (Fig. 3).

This paper is organized as follows. Section 2 contains the necessary background and basics of simplicial complexes and collapsibility, both from the classical simple homotopy type and the much more recent strong homotopy of Barmak and Minian [2,3]. Section 3 is devoted to both reviewing discrete Morse theory and introducing a generalized notion of critical object in this context. Here we develop a collapsing theorem for discrete Morse functions which is analogous to the classical result of Forman [10, Theorem 3.3].

Section 4 is the heart of the paper. In this section, we recall the definitions and basic properties of the simplicial L–S category and prove the simplicial L–S theorem in Theorem 4.5. The rest of the section is devoted to examples.

2 Fundamentals of Simplicial Complexes

In this section, we establish notation and terminology used throughout the body of the paper. We assume familiarity with the basics of simplicial complexes (but see [13] for a detailed exposition).

We use K or L to denote a simplicial complex, and $L \subseteq K$ to denote that L is a subcomplex of K. If a simplex τ is a face of a simplex σ, we write $\tau \leq \sigma$. We use $\sigma^{(i)}$

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to denote a simplex of dimension i, and we write $\tau < \sigma^{(i)}$ to denote any subsimplex of σ of dimension strictly less than i. If $\sigma, \tau \in K$ with $\tau < \sigma$, then τ is a *face* of σ and σ is a *coface* of τ. A simplex of K that is not properly contained in any other simplex of K is called a *facet* of K.

In order to discuss simple-homotopy theory, we also recall the classical notion of a collapse [5, Chap. 2, pp. 14–16].

Definition 2.1 Let K be a simplicial complex and suppose that there is a pair of simplices $\sigma^{(p)} < \tau^{(p+1)}$ in K such that σ is a face of τ and σ has no other cofaces. Such a pair $\{\sigma, \tau\}$ is called a *free pair*. Then $K - \{\sigma, \tau\}$ is a simplicial complex called an *elementary collapse* of K (see Fig. 2). The action of collapsing is denoted $K \searrow K - \{\sigma, \tau\}$. The inverse of an elementary collapse is called an elementary expansion.

More generally, K is said to *collapse* onto L if L can be obtained from K through a finite series of elementary collapses, denoted $K \searrow L$. In the case where $L = \{v\}$ is a single vertex, we say that K is *collapsible*.

It is important to point out that collapses are not simplicial maps in general. This suggests that it is natural to consider a special kind of collapse which is a simplicial map. The notion of strong collapse, introduced by Barmak and Minian [2,3], satisfies this requirement.

Now we introduce some basic subcomplexes related to a vertex. They play analogous role in the simplicial setting as the closed ball, sphere, and open ball play in the continuous approach.

Definition 2.2 Let K be a simplicial complex and let $v \in K$ be a vertex.

The *star of v in K*, denoted $\mathrm{st}(v)$, is the subcomplex of simplices $\sigma \in K$ such that $\sigma \cup \{v\} \in K$.

The *link of v in K*, denoted $\mathrm{lk}(v)$, is the subcomplex of $\mathrm{st}(v)$ of simplices which do not contain v.

The *open star of v in K* is $\mathrm{st}^o(v) := \mathrm{st}(v) - \mathrm{lk}(v)$. Note that $\mathrm{st}^o(v)$ is not a simplicial subcomplex, so it has to be considered in the geometric realization of K.

The *deletion of the vertex v* consists on the deletion of the open star of v in K. The resulting subcomplex is denoted by $K - \{v\}$.

Finally, given a simplicial complex K and a vertex $v \notin K$, the *cone vK* is the simplicial complex whose simplices are $\{v_0, \ldots, v_n\}$ and $\{v, v_0, \ldots, v_n\}$ where $\{v_0, \ldots, v_n\}$ is any simplex of K.

Now we recall the notions of contiguous maps and strong equivalence.

Definition 2.3 Let K, L be two simplicial complexes. Two simplicial maps $\varphi, \psi : K \to L$ are *contiguous*, denoted by $\varphi \sim_c \psi$, if for any simplex $\sigma = \{v_0, \ldots, v_p\}$ of K, the set of vertices

$$\varphi(\sigma) \cup \psi(\sigma) = \{\varphi(v_0), \ldots, \varphi(v_p), \psi(v_0), \ldots, \psi(v_p)\}$$

is a simplex of L. More generally, two simplicial maps $\varphi, \psi : K \to L$ are in the same *contiguity class*, denoted by $\varphi \sim \psi$, if there is a sequence of simplicial maps

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$\varphi_i : K \to L, i = 1, \ldots, m$, such that $\varphi = \varphi_1 \sim_c \varphi_2 \sim_c \cdots \sim_c \varphi_m = \psi$. A simplicial map $\varphi : K \to L$ is a *strong equivalence* is there exists a simplicial map $\psi : L \to K$ such that $\psi \circ \varphi \sim 1_K$ and $\varphi \circ \psi \sim 1_L$.

Barmak and Minian proved that the existence of a strong equivalence between two complexes is equivalent to having the same strong homotopy type (see [3, Cor. 2.12]).

Definition 2.4 Let K be a simplicial complex and suppose there exists a pair of vertices $v, v' \in K$ such that every maximal simplex containing v also contains v'. Then we say that v' *dominates* v and v is *dominated by* v'.

If v is dominated by v' then the inclusion $i : K - \{v\} \to K$ is a strong equivalence. Its homotopical inverse is the retraction $r : K \to K - \{v\}$ which is the identity on $K - \{v\}$ and such that $r(v) = v'$. Notice that r is a simplicial map. This retraction is called an *elementary strong collapse* from K to $K - \{v\}$, denoted by $K \searrow\!\!\searrow_e K - \{v\}$.

A *strong collapse* from K to L is a finite sequence of elementary strong collapses, denoted by $K \searrow\!\!\searrow L$. The inverse of an elementary strong collapse is called an *elementary strong expansion* and two complexes K and L *have the same strong homotopy type* if there is a sequence of elementary strong collapses and elementary strong expansions that transform K into L.

Equivalently, v is a dominated vertex if and only if its link is a cone [3, Rem. 2.3].

See Fig. 1 to illustrate the above notions.

It is interesting to point out that every elementary strong collapse can be decomposed into a finite sequence of elementary classical collapses, therefore the classical notion of collapse is weaker than the notion of strong collapse [3, Rem. 2.4].

3 Strong Discrete Morse Theory

We are now ready to introduce a key object of our study, that is, discrete Morse functions in the Forman's sense [10]. More precisely, we are interested in a generalized notion of critical object suitable for codifying the strong homotopy type of a complex.

Definition 3.1 Let K be a simplicial complex. A *discrete Morse function* $f : K \to \mathbb{R}$ is a function satisfying for any p-simplex $\sigma \in K$:

(M1) $\#\{\tau^{(p+1)} > \sigma : f(\tau) \leq f(\sigma)\} \leq 1$.
(M2) $\#\{\upsilon^{(p-1)} < \sigma : f(\upsilon) \geq f(\sigma)\} \leq 1$.

Definition 3.2 A *critical simplex* of f is a simplex σ satisfying:

(C1) $\#\{\tau^{(p+1)} > \sigma : f(\tau) \leq f(\sigma)\} = 0$.
(C2) $\#\{\upsilon^{(p-1)} < \sigma : f(\upsilon) \geq f(\sigma)\} = 0$.

If σ is a critical simplex, the number $f(\sigma) \in \mathbb{R}$ is called a *critical value*. Any simplex that is not critical is called a *regular simplex* while any output value of the discrete Morse function which is not a critical value is a *regular value*. The set of critical simplices of f is denoted by $\mathrm{crit}(f)$.

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Given any real number c, the *level subcomplex* of f at level c, $K(c)$, is the subcomplex of K consisting of all simplices τ with $f(\tau) \leq c$, as well as all of their faces, that is,

$$K(c) = \bigcup_{f(\tau)\leq c} \bigcup_{\sigma \leq \tau} \sigma.$$

A discrete Morse function f induces an ordered sequence of removals of critical simplices and elementary collapses by starting with the largest value of f and working down. For the discrete Morse function in Example 3.5, the largest value of f is 18, and since the vertex corresponding to 18 is part of a regular pair, removal of the vertex labeled 18 along with the corresponding edge labeled 17 is an elementary collapse. The next largest value is 16, corresponding to an edge/face pair with values 16 and 15, respectively. This too corresponds to an elementary collapse and hence is also removed, etc.

Any discrete Morse function induces the so-called gradient vector field defined as follows:

Definition 3.3 Let f be a discrete Morse function on K. The *induced gradient vector field* V_f or V when the context is clear, is the following set of pairs of simplices:

$$V_f := \left\{(\sigma^{(p)}, \tau^{(p+1)}) : \sigma < \tau, f(\sigma) \geq f(\tau)\right\}.$$

Note that critical simplices are easily identified in terms of the gradient field as precisely those simplices not contained in any pair in the gradient field.

A discrete Morse function is called *injective* if all its values are different except those on the simplices forming a pair in the gradient vector field, that is, if $f(\sigma) = f(\tau)$ then either $\sigma = \tau$ or $(\sigma, \tau) \in V_f$, for all simplices σ and τ.

Notice that to define a well-ordered sequence of removals of critical simplices and elementary collapses, the considered discrete Morse function needs to be injective (see Fig. 4). Therefore we will consider injective discrete Morse functions in all this paper.

Intuitively, a discrete Morse function defined on a simplicial complex K codifies one of two moves:

(1) Perform an elementary collapse in the current simplicial complex.
(2) Remove a facet from the current simplicial complex.

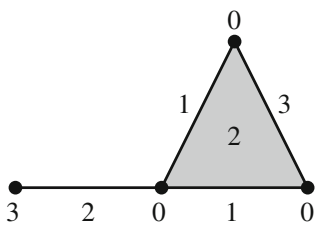

Fig. 4 A non-injective discrete Morse function

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A pair of simplices removed in (1) is a *free pair* while a simplex removed in (2) is a *critical simplex*. We propose adding a third basic move.

(0) Perform an elementary strong collapse.

With the addition of move (0), the pair removed in (1) becomes a *critical pair* (see Definition 3.8) and along with the set of simplices removed in (2), form the *critical objects*. Note that for a vertex/edge pair, a strong collapse is a regular collapse. The next several definitions are given in order to "detect" when we have performed an elementary strong collapse.

Specifically, we would like to view f as inducing a sequence of elementary strong collapses along with removal of critical simplices. However, there may be some elementary usual collapses which do not "fit" into a strong collapse. Such elementary classical collapses will be called *critical pairs* (see Definition 3.8). We thus desire to view f as inducing a sequence of strong collapses, removal of critical pairs, and removal of critical simplices. Part of the trick to doing this is to realize that the order of collapses induced by the discrete Morse function matters. Being more precise, if one begins performing elementary classical collapses in what looks like a strong collapse, but stops in the middle and start to collapse in a different part of the simplicial complex, those previous collapses become critical pairs. That is, critical pairs are detecting when a classical collapse, which is not part of a strong collapse, has been made. This may happen either due to combinatorial reasons (e.g. the non-existence of dominated vertices in the corresponding subcomplex) or to a bad choice of the values of the discrete Morse function (i.e. noise). This second option induces a bad ordering in the way the classical collapses are made inside a potential strong one. The key idea is that every elementary strong collapse should ideally be made as a uninterrupted sequence of classical collapses. So every time it is not made in this way, a new critical object appears, in terms of a critical pair corresponding to a classical collapse out of a strong one.

With this in mind, we make the following definitions.

Definition 3.4 Let $f\colon K \to \mathbb{R}$ be a discrete Morse function and V_f its induced gradient vector field. For each vertex/edge pair $(v, uv) \in V_f$ (where u and v are vertices of K), write $\mathrm{St}(v, u) := \mathrm{st}^o(v) \cap \mathrm{st}(u)$. Define

$$m_v^f := \min\left\{f(\tau) : f(\tau) > f(uv),\ \tau \in (K - \mathrm{St}(v, u)) \cup (\mathrm{crit}(f) \cap \mathrm{St}(v, u))\right\} \quad (2)$$

for the above set nonempty. Otherwise, define $m_v^f := \max\{f(\mathrm{St}(u, v))\}$. When the function f is clear from the context, we will sometimes write m_v.

The value m_v determines the smallest value of f greater than $f(uv)$ but outside of the $\mathrm{St}(u, v)$, i.e., outside of the "biggest possible" strong collapse induced by the vector (u, uv).

Example 3.5 We illustrate Definition 3.4. Let K be the simplicial complex with discrete Morse function f given in Fig. 5.

Observe that taking $v = f^{-1}(10)$ and $u = f^{-1}(0)$, we have $(v, uv) \in V_f$ and

$$\mathrm{St}(v, u) = f^{-1}(\{9, 10, 11, 12, 13, 14\}).$$

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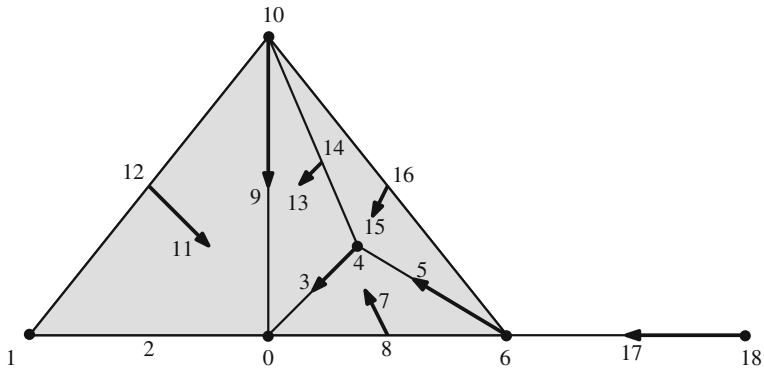

Fig. 5 A 2-dimensional simplicial complex with a discrete Morse function

Since $\mathrm{St}(v, u)$ in Example 3.5 does not contain any critical simplex, then determining m_v consists in finding the smallest value greater than $f(uv)$ outside of $\mathrm{St}(v, u)$. In this case, $m_v = 15$.

Example 3.6 The set

$$\left\{f(\tau) : f(\tau) > f(uv),\ \tau \in (K - \mathrm{St}(v, u)) \cup (\mathrm{crit}(f) \cap \mathrm{St}(v, u))\right\}$$

may sometimes be empty. Indeed, consider the vertex u and edge uv satisfying $f(u) = 18$ and $f(uv) = 17$ in Fig. 5. Then the set is clearly empty, and by definition, $m_v = 18$. In general, this set being empty corresponds to the discrete Morse function first performing a strong collapse(s) before detecting a critical object (see Definition 3.8).

Notice that both examples have the same induced gradient vector field and consequently, contain the same critical simplices in the usual Forman's sense. In other words, the information from the gradient vector field alone is not enough to detect whether or not a discrete Morse function induces a strong collapse in a row. For that, the order of the collapses must be taken into account. When a strong collapse is begun but only partially completed, this introduces a new kind of criticality, called a critical object, defined below.

Definition 3.7 Continuing with the notation used in Definition 3.4, define l_v^f as the largest regular value in $f(\mathrm{St}(v, u))$ such that

- $f(uv) \leq l_v^f \leq m_v^f$,
- every maximal regular simplex of $K(l_v) \cap \mathrm{St}(v, u)$ contains the vertex u.

We will write l_v for l_v^f when f is clear from the context.

Definition 3.8 Under the notation of Definition 3.4, the *strong subset* of v under f, denoted S_v^f, is given by

$$S_v^f := \{(\sigma, \tau) \in V_f : f(uv) \leq f(\tau) \leq l_v\} \tag{3}$$

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and the interval $I(S_v^f) = [f(uv), l_v]$ is called the *strong interval* of S_v^f. We will write S_v when f is clear from the context.

The elements of the set $C(V_f) := V_f - \bigcup_{v \in K} S_v^f$ are the *critical pairs* of f while each element in $\bigcup S_v^f$ is a *regular pair* of f. If (σ, τ) is a critical pair, the value $f(\tau)$ is a *critical value* of f or the *image of the critical pair* (σ, τ) under f.

A *critical object* is either a critical simplex (in the standard Forman's sense) or a critical pair. The set of all critical objects of f is denoted by $\mathrm{scrit}(f)$. In order to avoid confusion, we call the images of all critical objects (either in the Forman's sense or from a critical pair) *strong critical values*.

Lemma 3.9 *Let $v \neq v'$ be two vertices of K. If $I(S_v^f)$ and $I(S_{v'}^f)$ are strong intervals of f, then $I(S_v^f) \cap I(S_{v'}^f) = \emptyset$.*

Proof The values of the strong intervals are determined by its corresponding set of critical pairs, and these sets have no simplices in common. Furthermore, by definition, we have that $S_v \subseteq \mathrm{St}(v, u)$. The definition of m_v assures that the interval $[f(uv), l_v]$ does not contain any values under f outside of the images of simplices in S_v. □

The strong collapse and strong intervals are constructed precisely to detect strong collapses. Formally, we have

Lemma 3.10 *Suppose that $I(S_v^f) = [f(uv), l_v]$ is a strong interval. Then there exists $\epsilon > 0$ such that $K(l_v) \searrow_e K(f(uv) - \epsilon)$.*

Proof Let $t_v = \max\{t \in f(K) : t < f(uv)\}$ and $\epsilon > 0$ such that $t \leq f(uv) - \epsilon < f(uv)$. Since $l_v \leq m_v$, all the new simplices that are attached from $K(f(uv) - \epsilon)$ to obtain $K(l_v)$ are contained within $\mathrm{St}(v, u)$ and consequently in $st^o(v)$. By definition of l_v, u is contained in every maximal regular simplex of $K(l_v) \cap \mathrm{St}(v, u)$. Hence every maximal simplex in $K(l_v) - K(f(uv) - \epsilon)$ containing v also contains u and thus, v is a dominated vertex by u in $K(l_v)$, i.e., $K(l_v) \searrow_e K(f(uv) - \epsilon)$. □

Example 3.11 Now the function f of Example 3.5 will be slightly modified to create a new discrete Morse function g on K (Fig. 6). It is clear that $(v, uv) \in V_g$, but in this case $m_v = 11$.

We see that $l_v^f = 14$ while $l_v^g = 10$. Hence (by abuse of notation), (16, 15) is a critical pair under f while (14, 13), (16, 15) and (18, 17) are critical pairs under g. Notice that the strong intervals are $I(S_v^f) = [9, 14]$ and $I(S_v^g) = [9, 10]$, respectively.

Example 3.12 To further illustrate Definitions 3.7 and 3.8, we shall consider two different discrete Morse functions defined on a collapsible but non-strongly collapsible triangulation of the 2-disc. Notice that both of them have only a single critical value (in the Forman's sense), but very different numbers of critical objects, due to a bad election in the ordering of elementary collapses given by the values of g.

Let f be the discrete Morse function given in Fig. 7.

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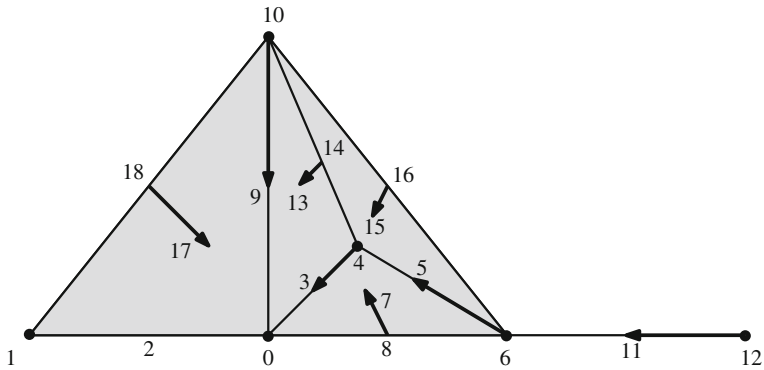

Fig. 6 The same 2-dimensional simplicial complex of Fig. 5 with another discrete Morse function

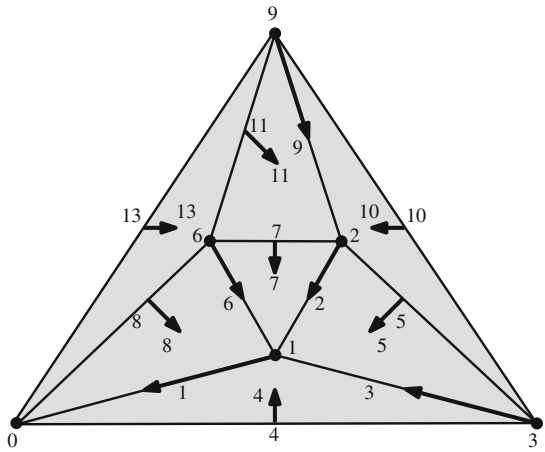

Fig. 7 A discrete Morse function defined on a non-strongly collapsible triangulation of the 2-disc

By abuse of notation, we refer to the simplices by their value under the discrete Morse function (the fact that pairs in V_f are given the same label should not cause confusion). For each of the pairs $(9, 9), (6, 6), (3, 3), (2, 2), (1, 1) \in V_f$, we have the corresponding values $l_9 = 11, l_6 = 8, l_3 = 5, l_2 = 2$, and $l_1 = 1$. The corresponding strong collapses under the indicated vertices are given by

$$\begin{aligned}
S_9^f &= \{(9, 9), (10, 10), (11, 11)\},\\
S_6^f &= \{(6, 6), (7, 7), (8, 8)\},\\
S_3^f &= \{(3, 3), (4, 4), (5, 5)\},\\
S_2^f &= \{(2, 2)\} \text{ and}\\
S_1^f &= \{(1, 1)\}.
\end{aligned}$$

Thus we obtain the following strong intervals:

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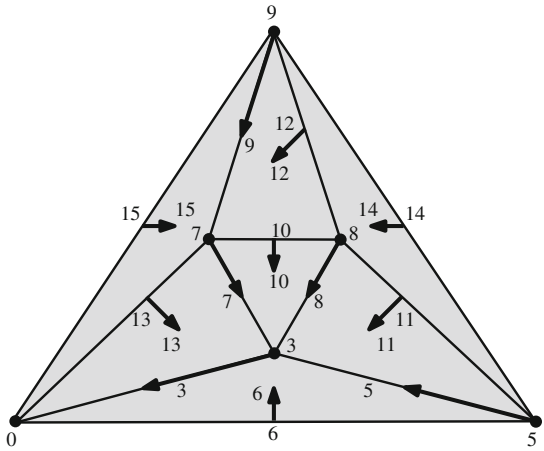

Fig. 8 A different discrete Morse function defined on a non-strongly collapsible triangulation of the 2-disc

$$I(S_9^f) = [9, 11], \quad I(S_6^f) = [6, 8], \quad I(S_3^f) = [3, 5],$$
$$I(S_2^f) = [2, 2] = \{2\} \quad \text{and} \quad I(S_1^f) = [1, 1] = \{1\}.$$

Hence there is a single critical pair, namely, (13, 13), so that

$$\text{scrit}(f) = \{0, (13, 13)\}.$$

It is worthwhile to mention that this discrete Morse function can be considered as optimal in the sense that it minimizes the number of critical objects, as we will see in Theorem 4.5.

Now let g be the discrete Morse function given in Fig. 8.

Now we consider the pairs $(9, 9), (8, 8), (7, 7), (5, 5), (3, 3) \in V_g$ and obtain corresponding values $l_9 = 9, l_8 = 8, l_7 = 7, l_5 = 6$ and $l_3 = 3$. The corresponding strong collapses are then given by

$$S_9^g = \{(9, 9)\}, \quad S_8^g = \{(8, 8)\}, \quad S_7^g = \{(7, 7)\},$$
$$S_5^g = \{(5, 5), (6, 6)\} \quad \text{and} \quad S_3^g = \{(3, 3)\}.$$

It follows that the strong intervals are:

$$I(S_9^g) = [9, 9] = \{9\}, \quad I(S_8^g) = [8, 8] = \{8\}, \quad I(S_7^g) = [7, 7] = \{7\},$$
$$I(S_5^g) = [5, 6] \quad \text{and} \quad I(S_3^g) = [3, 3] = \{3\}.$$

We conclude that

$$\text{scrit}(g) = \{0, (15, 15), (13, 13), (14, 14), (10, 10), (12, 12), (11, 11)\}$$

for a total of seven critical objects.

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At this point an interesting question arises: given two discrete Morse functions on a complex with the same induced gradient field, how do the corresponding numbers of critical objects differ? Being more precise, given a simplicial complex K and a discrete Morse function f on K, we define the following numbers:

$$M(f) = \max\{\#\operatorname{scrit}(g) : g \text{ is a discrete Morse function on } K \text{ such that } V_g = V_f\},$$
$$m(f) = \min\{\#\operatorname{scrit}(g) : g \text{ is a discrete Morse function on } K \text{ such that } V_g = V_f\}.$$

We are interested to know how far away are $M(f)$ and $m(f)$. As we shall see in the next example, they can be arbitrarily distant, restricted to the same triangulation of K.

Example 3.13 Starting from the function f defined in Example 3.12, we construct a family of functions f_i (showed in Fig. 9) with $2 \leq i \leq 7$ on the same complex such that $V_{f_i} = V_f$ increasing the number of critical objects as i goes up.

We emphasize that all these functions have a unique critical vertex (in the Forman's sense) and they have as many critical pairs as the index i.

The following theorem is the analogue of Forman's classical result of [10, Theorem 3.3] and also analogous to the classical result in the smooth setting establishing homotopy equivalence of sublevel sets by assuming non-existence of critical values.

Theorem 3.14 *Let f be a discrete Morse function on K and suppose that f has no strong critical values on $[a, b]$. Then there exists a value $a' \leq a$ such that $K(b) \searrow\!\!\searrow K(a')$. The value a' is maximum in the sense that if a'' is any other value such that $K(b) \searrow\!\!\searrow K(a'')$, then $K(a'') \subseteq K(a')$.*

In particular, if $I(S_v^f) = [f(uv), l_v]$ is a strong interval for some vertex $v \in K$, then $K(l_v) \searrow\!\!\searrow_e K(f(uv))$.

Proof Since the image of every simplex of a discrete Morse function is either a critical value or it is contained in a strong interval, then the case where $[a, b]$ does not intersect a strong interval implies that $[a, b] \cap f(K) = \emptyset$ so that $K(b) = K(a)$. Hence, suppose that $[a, b]$ has nonempty intersection with at least one strong interval. By Lemma 3.9, all strong intervals are disjoint, and hence, we may partition $[a, b]$ into subintervals such that each one of them intersects exactly one strong interval. Let us thus suppose that $I(S_v) = [f(uv), l_v] \subseteq [a, b]$ is the unique strong interval intersecting $[a, b]$. If $l_v < b$, then the interval $(l_v, b] \cap f(K) = \emptyset$. For otherwise, $(l_v, b]$ would contain either a strong critical value, contradicting the hypothesis, or it contains an element of another strong interval, contradicting our assumption that $[a, b]$ intersected only one strong interval. By Lemma 3.10, $K(b) = K(l_v) \searrow\!\!\searrow_e K(f(uv) - \epsilon)$. If $l_v \geq b$, then u still dominates v in $K(b) \subseteq K(l_v)$. Hence in either case, $K(b) \searrow\!\!\searrow_e K(f(uv) - \epsilon)$. Finally, if $a \geq f(uv)$, choose $a' := f(uv) - \epsilon$. Otherwise, if $a < f(uv)$, ϵ may be chosen so that $a' = f(uv) - \epsilon := a$. Thus the result. □

4 Simplicial Lusternik–Schnirelmann Category

In this section, we recall the fundamental definitions and results found in [9].

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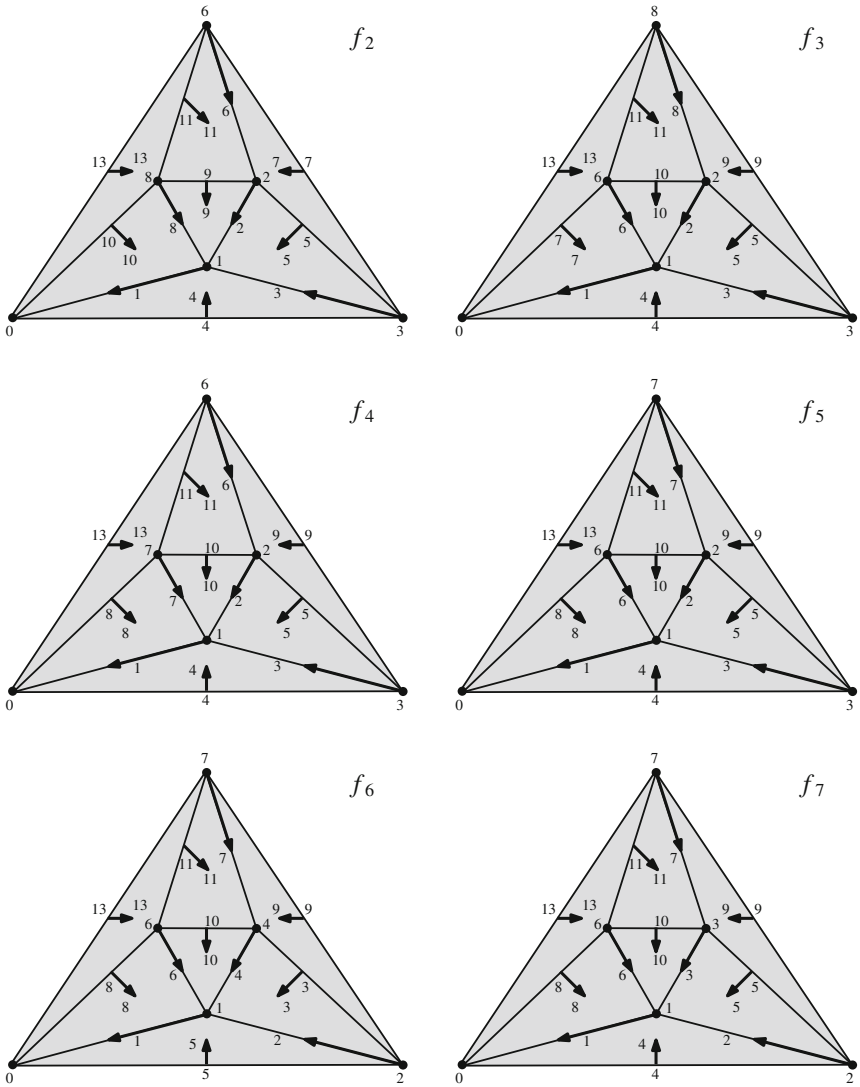

Fig. 9 Family of discrete Morse functions on a non-strongly collapsible triangulation of the 2-disc with the same gradient field and increasing criticality

Definition 4.1 Let K, L be simplicial complexes. We say that two simplicial maps $\phi, \psi\colon K \to L$ are *contiguous*, denoted $\phi \sim_{\mathrm{c}} \psi$, if for any simplex $\sigma \in K$, we have that $\phi(\sigma) \cup \psi(\sigma)$ is a simplex of L. Because this relation is reflexive and symmetric but not transitive, we say that ϕ and ψ are in the same *contiguity class*, denoted $\phi \sim \psi$, if there is a sequence $\phi = \phi_0 \sim_{\mathrm{c}} \phi_1 \sim_{\mathrm{c}} \ldots \sim_{\mathrm{c}} \phi_n = \psi$ of contiguous simplicial maps $\phi_i\colon K \to L, 0 \le i \le n$. A map $\phi\colon K \to L$ is a *strong equivalence* if there exists $\psi\colon L \to K$ such that $\psi \circ \phi \sim \mathrm{id}_K$ and $\phi \circ \psi \sim \mathrm{id}_L$. In this case, we say that K and L are *strongly equivalent*, denoted by $K \sim L$.

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There is a nice link between strong equivalences and strong collapses.

Theorem 4.2 [3, Cor. 2.12] *Two complexes K and L have the same strong homotopy type if and only if $K \sim L$.*

Definition 4.3 [9] Let K be a simplicial complex. We say that the subcomplex $U \subseteq K$ is *categorical* in K if there exists a vertex $v \in K$ such that the inclusion $i : U \to K$ and the constant map $c_v : U \to K$ are in the same contiguity class. The *simplicial L–S category*, denoted $\mathrm{scat}(K)$, of the simplicial complex K, is the least integer $m \geq 0$ such that K can be covered by $m + 1$ categorical subcomplexes.

One of the basic results of classic L–S theory states that the L–S category is homotopy invariant. The next result shows that simplicial L–S category satisfies the analogous property in the discrete setting.

Theorem 4.4 [9, Thm. 3.4] *Let $K \sim L$ be two strongly equivalent complexes. Then* $\mathrm{scat}(K) = \mathrm{scat}(L)$.

We refer the interested reader to the papers [8,9] for a detailed study of this topic.

4.1 Simplicial Lusternik–Schnirelmann Theorem

Our main result is the following simplicial version of the L–S theorem:

Theorem 4.5 *Let $f : K \to \mathbb{R}$ be a discrete Morse function. Then*

$$\mathrm{scat}(K) + 1 \leq \#\,\mathrm{scrit}(f). \tag{4}$$

Proof For any natural number $n < \mathrm{scat}(K)$, define

$$c_n := \min\{a \in \mathbb{R} : \mathrm{scat}(K(a)) \geq n - 1\}.$$

We claim that c_n is a strong critical value of f. If c_n is a regular value, then it is either contained in a strong interval or it is not. If c_n is contained in a strong interval $I(S_v) = [f(uv), l_v]$, then by Theorem 3.14, we have $K(c_n) \searrow\!\!\searrow_e K(f(uv) - \epsilon)$ and that $\mathrm{scat}(K(f(uv) - \epsilon)) = \mathrm{scat}(K(c_n))$ by Theorem 4.4, contradicting the minimality of c_n. Otherwise, c_n is outside a strong interval. But as observed in Theorem 3.14, if c_n is outside a strong interval and not a strong critical value, then c_n is not in the image of f, so there exists $\epsilon > 0$ such that $K(c_n) = K(c_n - \epsilon)$. But $c_n > c_n - \epsilon$ and c_n was the minimum value such that $\mathrm{scat}(K(c_n)) = n - 1$, which is a contradiction. Thus each c_n is a strong critical value of f.

We now prove by induction on n that $K(c_n)$ must contain at least n critical objects. By the well-ordering principle, the set $\mathrm{Im}(f)$ has a minimum, say $f(v) = 0$ for some 0-simplex $v \in K$. For $n = 1$, $c_1 = 0$ so that $K(c_1)$ contains 1 critical object. For the inductive hypothesis, suppose that $K(c_n)$ contains at least n strong critical objects. Since all the strong critical values of f are distinct, $c_n < c_{n+1}$ so that there is at least one new critical object in $f^{-1}(c_{n+1})$. Thus $K(c_{n+1})$ contains at least $n+1$ critical objects. Hence if $c_1 < c_2 < \cdots < c_{\mathrm{scat}(K)+1}$ are the critical objects, then $K(c_{\mathrm{scat}(K)+1}) \subseteq K$ contains at least $\mathrm{scat}(K) + 1$ critical objects. Thus $\mathrm{scat}(K) + 1 \leq \#\,\mathrm{scrit}(f)$. □

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Example 4.6 We give an example where the upper bound of Theorem 4.5 is attained. Let A be the several times considered non-strongly collapsible triangulation of the 2-disc with the discrete Morse function f given in Example 3.12. This satisfies $\#\operatorname{scrit}(f) = 2$. Since A does not have dominating vertices, then $\operatorname{scat}(A) > 0$, whence $\operatorname{scat}(A) + 1 = \#\operatorname{scrit}(f) = 2$.

Notice that just adding one simplex to A, the above equality turns into a strict inequality and thus, the number of critical objects may increase while the simplicial category keeps the same. To see this, let us consider A' as the clique complex of A, that is, we add to A the triangle generated by its three bounding vertices. This is a simplicial 2-sphere, so by means of Morse inequalities, it follows that every discrete Morse function f defined on A' has at least two critical simplices: one critical vertex (global minimum) and a critical triangle (global maximum). In addition, since A' is a simplicial 2-sphere, then it does not contain any dominated vertex. Moreover, after removing the critical triangle (and hence obtaining a complex isomorphic to A) no new dominated vertices appear, so at least one critical pair arises in order to collapse A to a subcomplex containing dominated vertices. Hence, we conclude that any discrete Morse function $f\colon A' \to \mathbb{R}$ must have at least three critical objects, which could be considered optimal in the sense that it minimizes the number of critical objects. Furthermore, it is easy to cover A' with two categorical sets so that $\operatorname{scat}(A') = 1$. Hence $\operatorname{scat}(A') + 1 = 2 < 3 \leq \#\operatorname{scrit}(f)$.

Taking into account the above example a natural question arises: given a discrete Morse function f on a complex K, how far away is the number of critical objects of f from $\operatorname{scat}(K) + 1$? Let us consider

$$m_K = \min\{\#\operatorname{scrit}(f) : f \text{ is a discrete Morse function on } K\}.$$

The next example illustrates how the distance between m_K and $\operatorname{scat}(K) + 1$ is strongly linked with the combinatorial structure of K.

Example 4.7 Let us consider the following irreducible triangulation of the Möbius band M_2 [4] where a discrete Morse function g is defined (see Fig. 10). Notice that g does not have any critical pair, actually it only has two critical simplices (one vertex and one edge) in the Forman's sense. Therefore, $m_{M_2} = \#\operatorname{scrit}(g) = 2$. It is easy to find two categorical subcomplexes covering M_2, so we conclude that $\operatorname{scat}(M_2) = 1$ and thus,

$$m_{M_2} - (\operatorname{scat}(M_2) + 1) = 0.$$

So the lower bound given by Theorem 4.5 is attained.

Now, let us consider another irreducible triangulation of the Möbius band M_4 [4] where a discrete Morse function h is defined (see Fig. 11). Notice that h has two critical pairs and it has also two critical simplices in the Forman's sense. Since this triangulation does not contain any dominated vertex and the same fact happens after removing a pair edge-triangle, then $m_{M_4} = \#\operatorname{scrit}(h) = 4$. Again, it is easy to find two categorical subcomplexes covering M_4, so we conclude that $\operatorname{scat}(M_4) = 1$ and thus,

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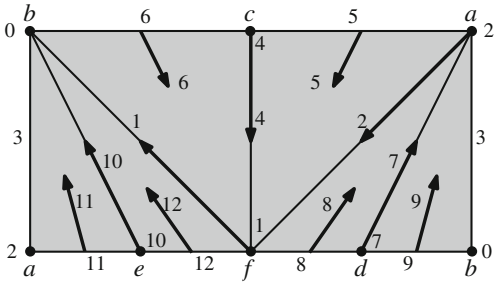

Fig. 10 A discrete Morse function on the triangulation M_2 of the Möbius band

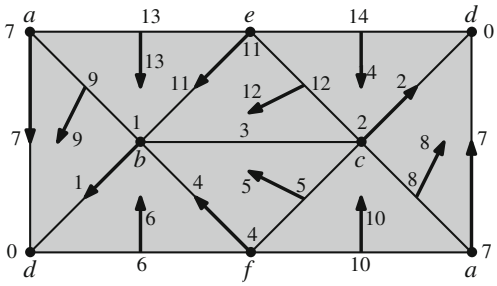

Fig. 11 A discrete Morse function on the triangulation M_4 of the Möbius band

$$m_{M_4} - (\mathrm{scat}(M_4) + 1) = 2.$$

We guess that the distance between m_K and $\mathrm{scat}(K) + 1$ can be arbitrarily large. It depends on the combinatorial structure of the complex (non-existence of dominated vertices) and the number of simplices.

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